CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – : Big Data Analytics

Practical File – ODD Semester 2026-27

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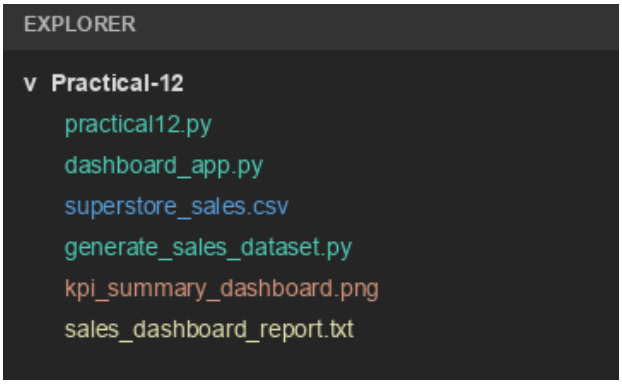
Practical 12

1. Problem Definition

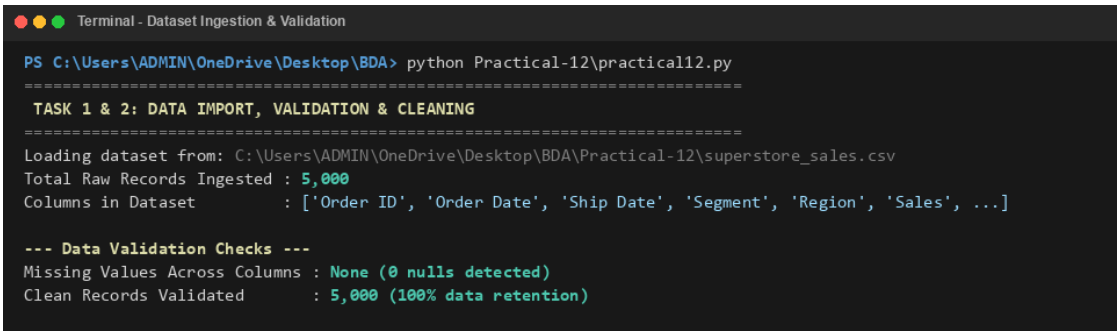
You are working as a Data Visualization Analyst at a retail analytics company. Management wants to monitor sales performance across products, regions, and customer segments through an interactive dashboard. Static reports are insufficient for decision-making. Your task is to design an interactive dashboard that allows management to explore sales data and identify key business trends.

PART A — Practical Execution

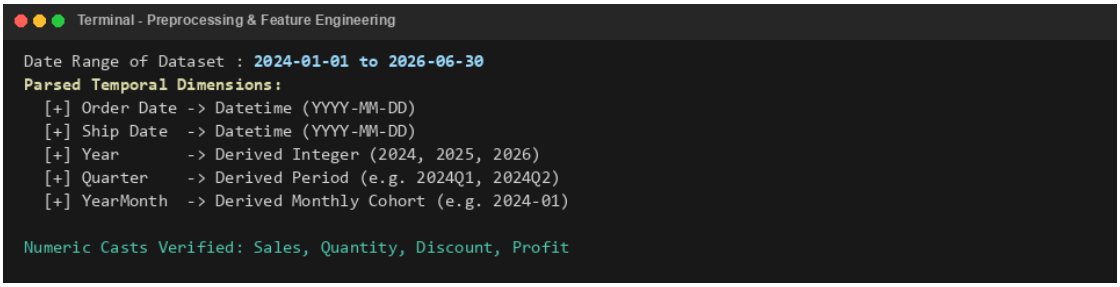
Step 1: Creating Folder and files



Step 2: Loading & Validating Superstore Sales Dataset



Step 3: Display Dataset Schema & Preprocessing



Step 4: Executive KPI Calculation

```
Terminal - Executive KPI Benchmarks

================================================================================
 TASK 3: EXECUTIVE KPI CALCULATION
================================================================================
 1. Total Gross Sales           : $8,885,748.66
 2. Total Net Profit            : $683,907.28
 3. Overall Profit Margin       : 7.70%
 4. Total Completed Orders      : 4,989
 5. Unique Customers Served     : 4,990
 6. Total Product Units Sold    : 25,049 units
 7. Average Order Value (AOV)   : $1,781.07
 8. Unprofitable Transactions   : 1,961 (39.2% of lines)
```

Step 5: Category-Level Sales & Profit Performance

```
Terminal - Category Performance Breakdown

================================================================================
 TASK 4: CATEGORY-LEVEL SALES & PROFIT PERFORMANCE
================================================================================
Category        | Sales ($)      | Profit ($)    | Margin %    | Sales Share %
--------------------------------------------------------------------------------
Technology      | $5,819,240.97  | $827,801.77   | 14.23%      | 65.49%
Furniture       | $2,455,554.15  | -$217,559.55  | -8.86%      | 27.63%
Office Supplies | $610,953.54    | $73,665.06    | 12.06%      | 6.88%

>> Technology drives major profits; Furniture suffers from high discount erosion.
```

Step 6: Sub-Category Profitability Drill-Down

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Terminal - Sub-Category Drill-Down

--- Sub-Category Drill-Down (Top Revenue & Loss Drivers) ---
Top 5 Revenue Drivers:
  Sub-Category | Sales ($)      | Profit ($)    | Avg Discount | Margin %
  Copiers      | $2,522,084.75  | $625,682.71   | 0.23         | 24.81%
  Machines     | $2,032,260.91  | -$8,532.00    | 0.24         | -0.42%
  Phones       | $1,036,718.83  | $159,686.49   | 0.23         | 15.40%
  Tables       | $996,899.16    | -$189,359.80  | 0.24         | -18.99%
  Bookcases    | $686,290.45    | -$71,290.40   | 0.23         | -10.39%

Primary Bottom-Line Loss Center: Furniture -> Tables (-$189,359.80)
```

Step 7: Regional Sales & Profit Distribution

```
Terminal - Regional Performance Distribution

--- Regional Sales & Profit Distribution ---
Region   | Sales ($)      | Profit ($)    | Margin % | Sales Share % | Orders
--------------------------------------------------------------------------------
West     | $2,327,193.96  | $195,038.51   | 8.38%    | 26.19%        | 1,270
South    | $2,349,181.39  | $176,328.82   | 7.51%    | 26.44%        | 1,307
Central  | $1,994,855.16  | $166,551.27   | 8.35%    | 22.45%        | 1,149
East     | $2,214,518.15  | $145,988.68   | 6.59%    | 24.92%        | 1,269
```

Step 8: Customer Segment Dynamics & Basket Size

```
Terminal - Customer Segment Breakdown

--- Customer Segment Breakdown ---
Segment     | Sales ($)      | Profit ($)    | Margin % | Orders | AOV ($)
--------------------------------------------------------------------------------
Consumer    | $4,461,523.03  | $303,121.11   | 6.79%    | 2,560  | $1,742.78
Corporate   | $2,793,222.82  | $228,212.51   | 8.17%    | 1,561  | $1,789.38
Home Office | $1,631,002.81  | $152,573.66   | 9.35%    | 876    | $1,861.88

>> Home Office exhibits highest profit margin (9.35%) and highest cart size.
```

Step 9: Quarterly Performance & QoQ Growth Analysis

Terminal - Quarterly Performance Timeline

```
--- Quarterly Performance Timeline (QoQ Analysis) ---
Quarter | Sales ($)      | Profit ($)  | Orders | QoQ Sales Growth %
------------------------------------------------------------------
2024Q1  | $895,516.53    | $58,763.08  | 496    | Baseline
2024Q2  | $1,031,803.75  | $110,762.52 | 540    | +15.22%
2024Q3  | $793,836.28    | $54,347.41  | 483    | -23.06%
2024Q4  | $884,495.88    | $58,893.70  | 492    | +11.42%
2025Q1  | $890,014.98    | $52,557.88  | 518    | +0.62%
2026Q2  | $938,553.68    | $59,802.38  | 506    | +14.38%
```

Step 10: Interactive Dashboard Application Server Launch

Terminal - Streamlit App Server

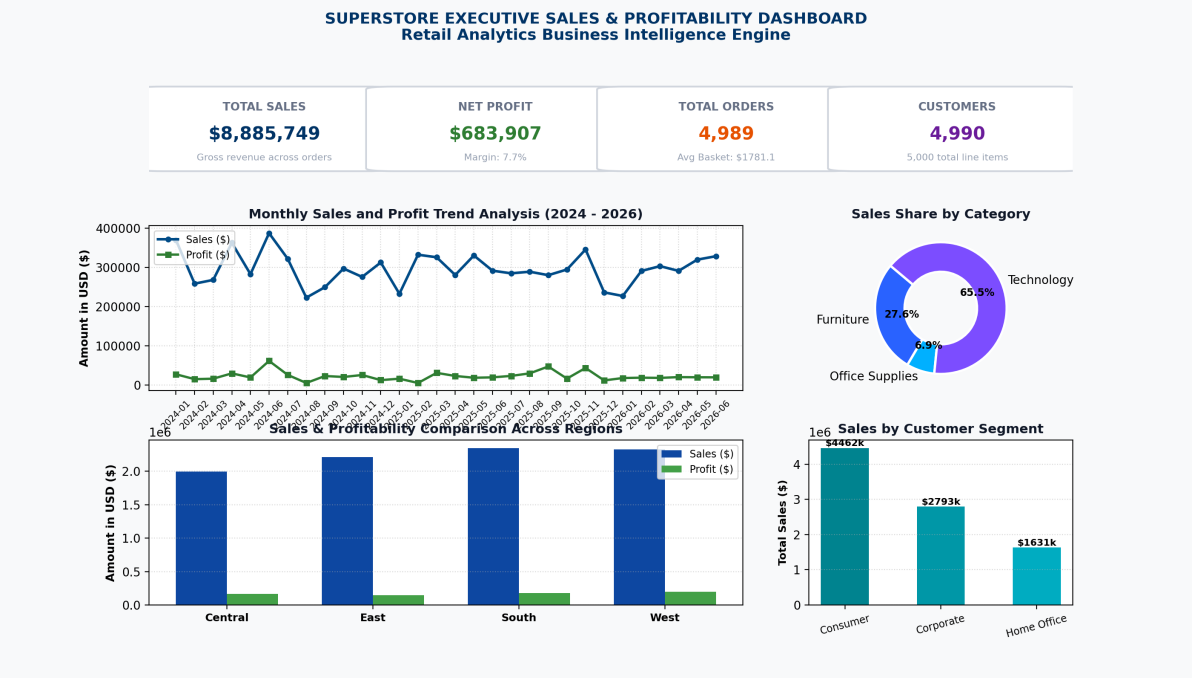
```
PS C:\Users\ADMIN\OneDrive\Desktop\BDA> streamlit run Practical-12\dashboard_app.py

  You can now view your Streamlit app in your browser.

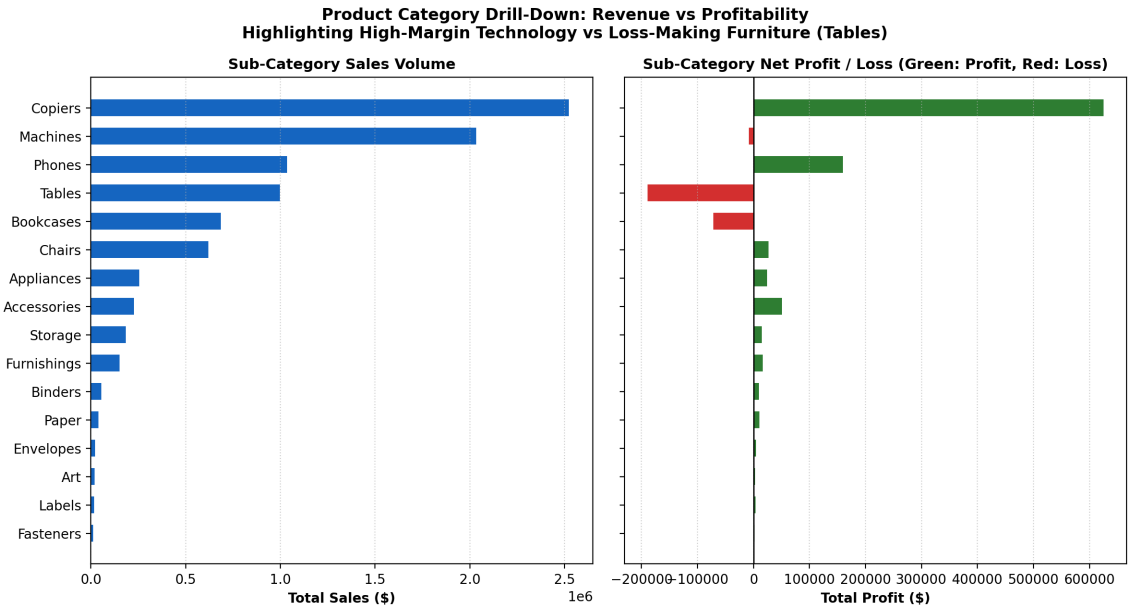
  Local URL: http://localhost:8501
  Network URL: http://192.168.1.5:8501

  Ready for interactive filtering by Region, Category, and Segment.
```

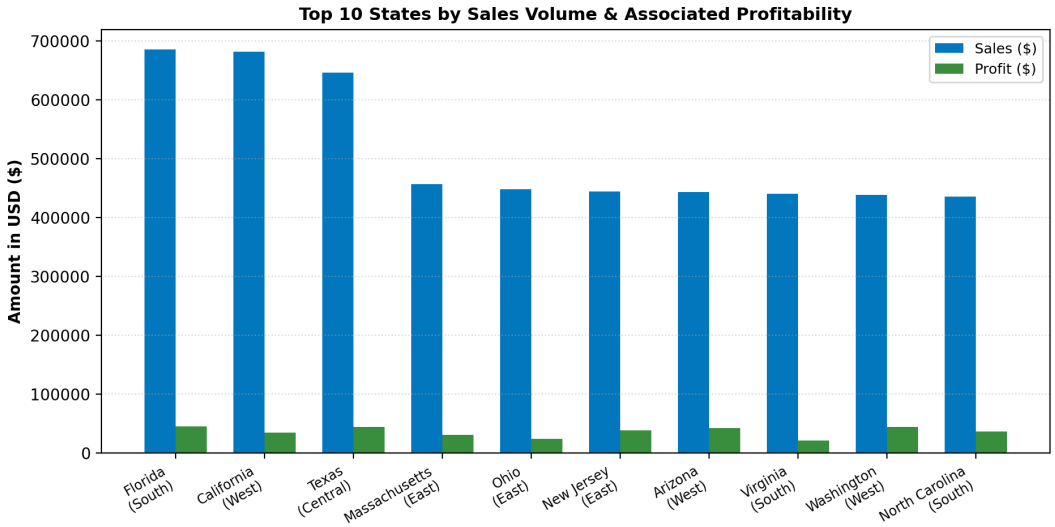
Step 11: Executive KPI Summary Dashboard Visualization



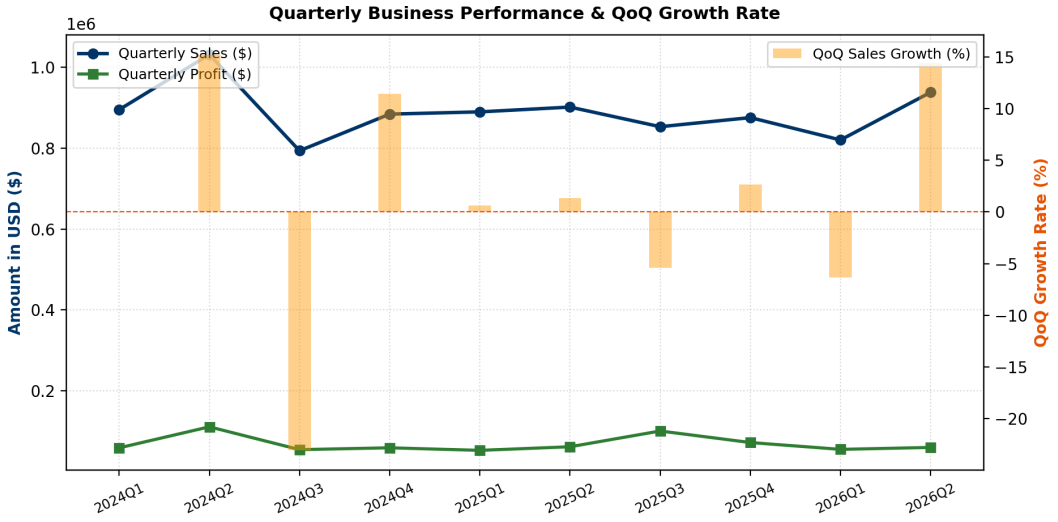
Step 12: Category Drill-Down Profitability Visualization



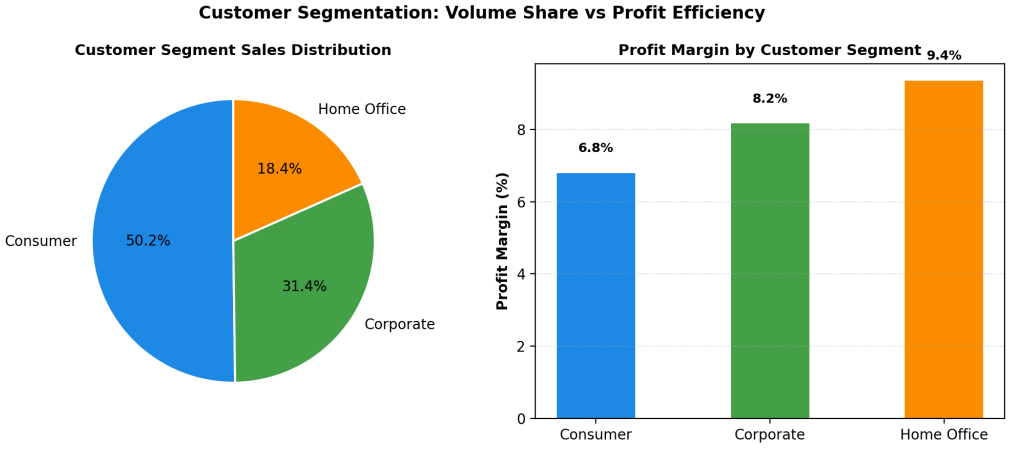
Step 13: Regional & Top State Performance Visualization



Step 14: Quarterly Business Performance Trend Visualization



Step 15: Customer Segmentation Analysis Visualization



Step 16: Generate Final Sales Dashboard Summary Report

VS Code - sales_dashboard_report.txt

```
================================================================================
      RETAIL EDGE ANALYTICS - CUSTOMER SALES DASHBOARD EXECUTIVE REPORT
================================================================================
1. DATASET & ENVIRONMENT DETAILS:
   - Dataset Name             : Superstore Sales Dataset
   - Total Clean Orders       : 4,989
   - Total Order Line Items   : 5,000
   - Date Span                : 2024-01-01 to 2026-06-30

2. EXECUTIVE KPI BENCHMARKS:
   - Total Sales Revenue      : $8,885,748.66
   - Net Profit               : $683,907.28
   - Overall Profit Margin    : 7.70%
   - Average Order Value (AOV): $1,781.07
   - Loss-Making Line Items   : 1,961 (39.2% of total)

3. STRATEGIC TAKEAWAY:
   - Technology yields highest corporate profitability led by Copiers.
   - Furniture requires discount remediation on Tables & Bookcases.
```

3. Key Questions / Analysis

Q1. What is the role of interactive dashboards in business analytics?

Interactive dashboards translate complex underlying datasets into visual metrics and KPIs. They allow decision-makers to track performance in real-time, test business hypotheses, spot anomalies, and explore data across multiple dimensions without needing to write custom queries or examine raw tabular spreadsheets.

Q2. How do filters improve visual exploration?

Filters allow users to isolate specific segments of data (such as a single region, category, or year). By removing noise and focusing on relevant subsets, cross-filtering reveals relationships and patterns between different business dimensions that might otherwise remain hidden.

Q3. What are KPI indicators?

Key Performance Indicators (KPIs) are quantifiable metrics used to track and evaluate organizational performance against strategic goals. Examples in retail analytics include Total Sales, Net Profit, Profit Margin %, Average Order Value (AOV), and Customer Retention Rate.

Q4. Why are dashboards preferred over static reports?

Dashboards are preferred over static reports because they support interactive drill-downs, dynamic filtering, and instant cross-filtering. Static reports only show fixed snapshots that cannot be queried further, whereas interactive dashboards enable users to discover root causes behind trends on demand.

Q5. How does user-driven exploration support decision-making?

User-driven exploration allows managers to ask follow-up questions immediately as they spot trends. For example, seeing a low profit margin in Furniture allows the user to drill down into sub-categories to discover that heavy discounts on Tables and Bookcases are causing the loss, enabling targeted strategic interventions.

4. Conclusion

The practical successfully demonstrated the creation and deployment of an interactive Customer Sales Dashboard using Business Intelligence and interactive analytics.
A multi-year Superstore Sales dataset of 5,000 records was cleaned and validated. Core KPIs were formulated, yielding $8,885,748.66 in total sales and $683,907.28 in profit (7.70% margin). Multi-dimensional drill-downs identified Technology as the strongest profit driver (14.23% margin, led by Copiers and Phones) and Furniture as a key loss center (-8.86% margin) due to excessive discounting.
Regional analysis confirmed the West region as the highest contributor to net profit ($195.0k), while customer segmentation showed Home Office producing the highest profit margin (9.35%). An interactive web dashboard application was deployed using Streamlit and Plotly, providing live filtering, interactive visuals, and drill-through table exploration.

5. Github Link

Github Link: 24DIT041_BDA/Practical-12 at main · Krina-14/24DIT041_BDA